Controller Error Solutions

Run way

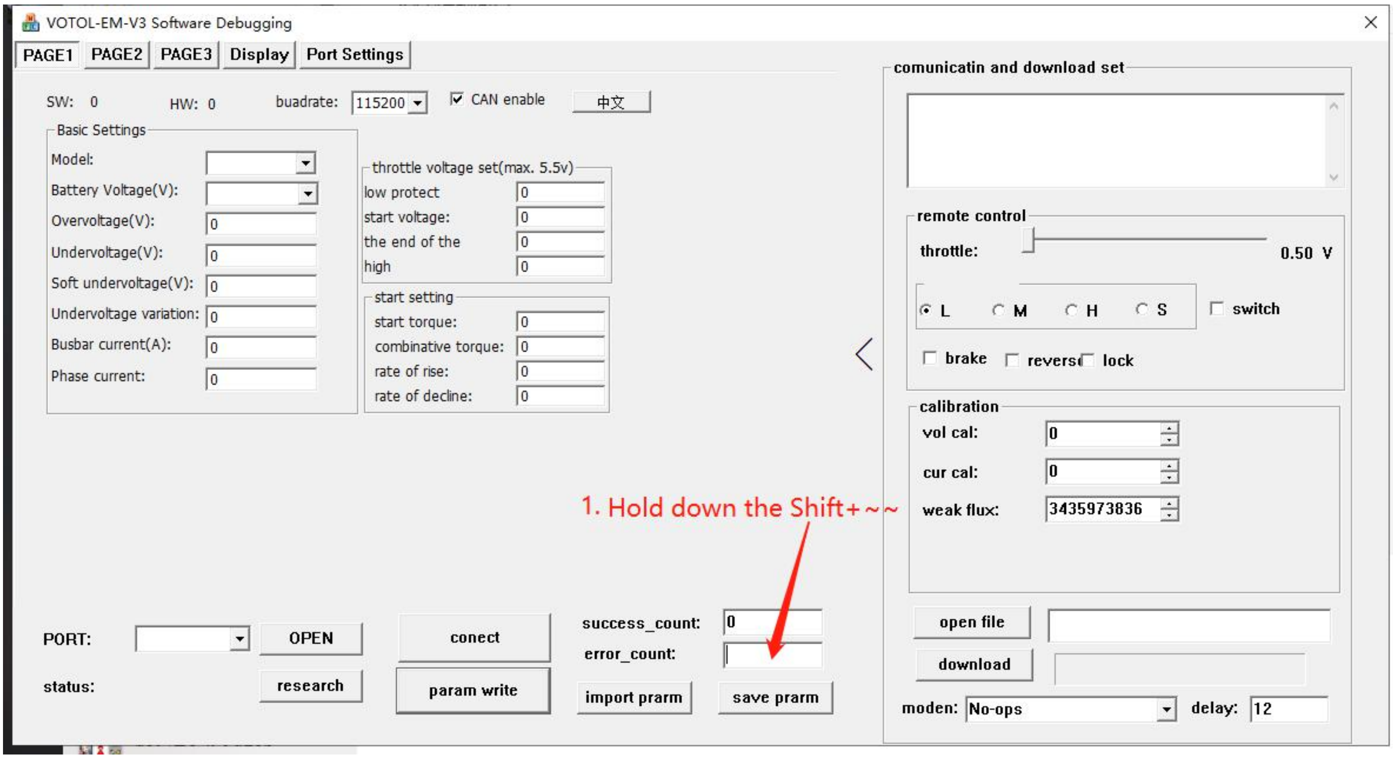
Problem: Customer feedback the controller is powered on and turn the handle,Rotate to the maximum speed, loose the handle and cannot stop

Reason:

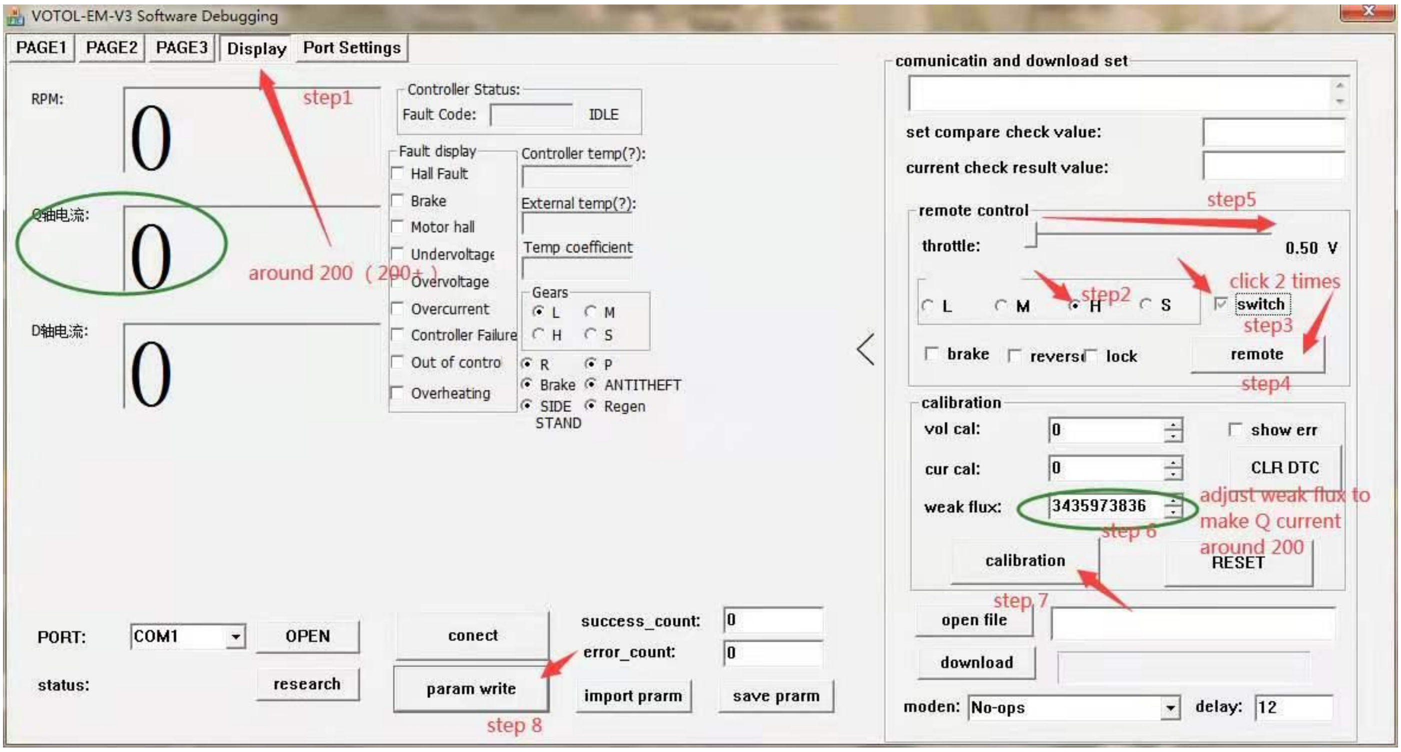
1. May be caused by mismatched controller configuration before leaving the factory
2. Procedural issues
3. The customer modified the parameters
4. Firmware need to be updated

Solutions:

1. Connect the controller to the host computer and enter the advanced settings

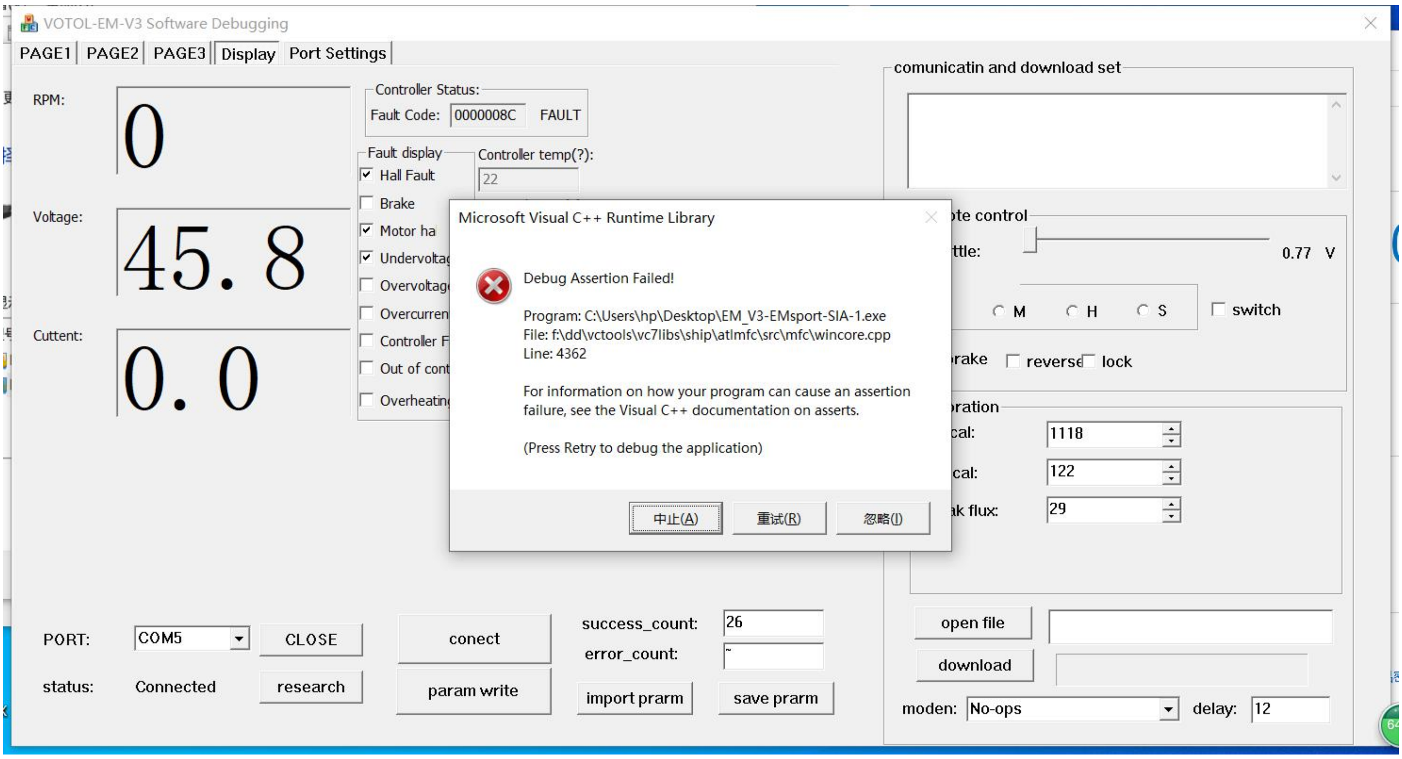


2. Enter the remote mode, adjust the Q-axis current to 200+ (see the figure for details)



The host computer reports an error

Problem: host computer reports an error



Reason: Controller software problem, temporarily unable to repair

Solutions:
Do not enter the advanced settings of the controller in the Display interface, otherwise the controller will report the above error

Discontinued device manager port

Problem: An exclamation mark appears on the device manager port!

Reason: The VOTOL controller driver is out of date,only this version of the driver can be installed PL2303_Prolific_DriverInstaller_v110
Solutions: Refer to the instructions for installing the serial port in 1.1 in the LAN VOTOL controller manual V3.1 to solve the problem.

Motor jitter

Problem: The motor jitters at high speed
Reason:

1. The controller does not actually match the motor when it leaves the factory
2. Individual Hall angle differences of motors
3. Solutions: 1. Connect to the host computer, in the setting interface 2

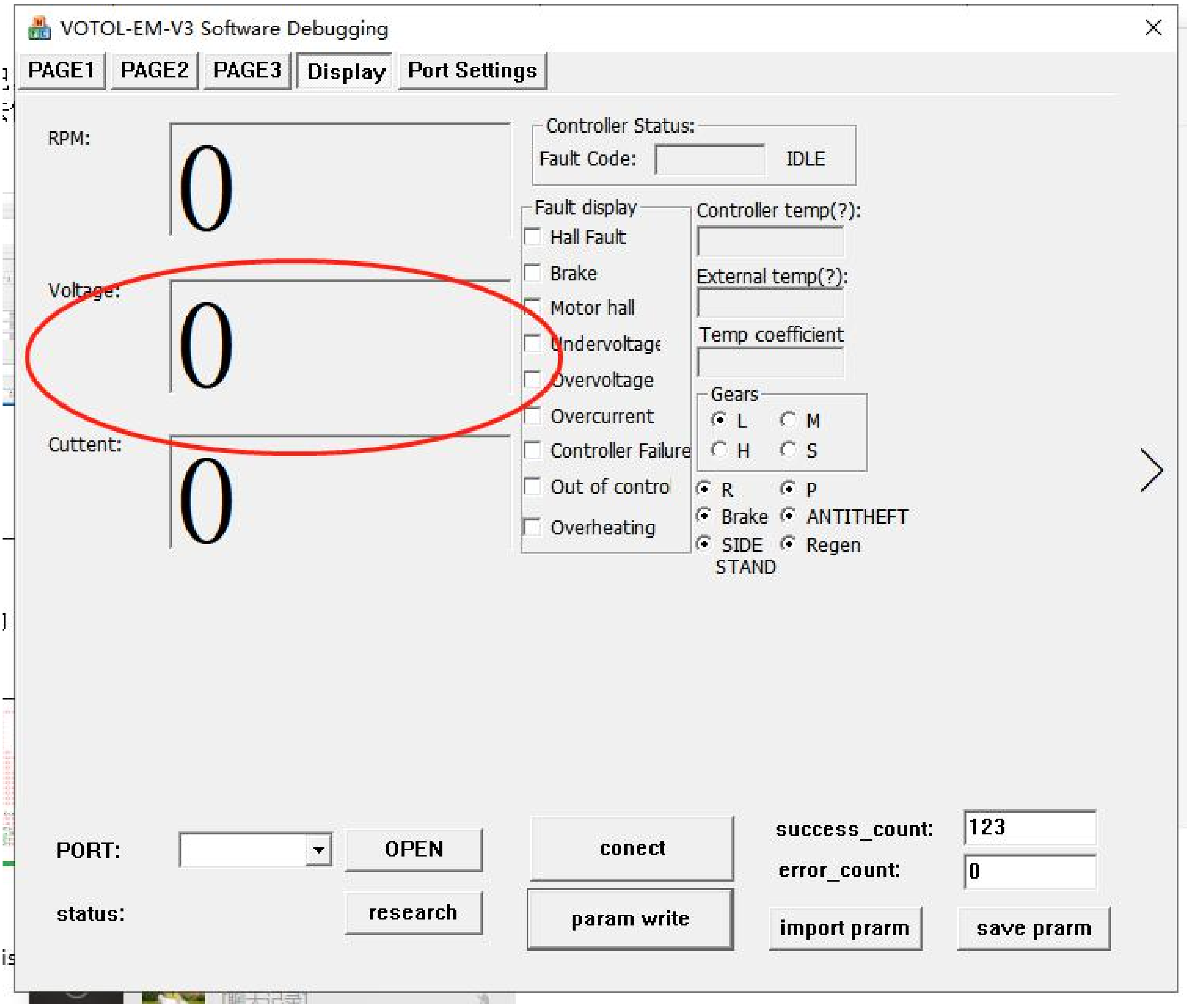
Controller fault code

```
#define E_BRAKE_ON              (u32)0x001
#define OVER_CURRENT            (u32)0x02
#define UNDER_VOLTAGE           (u32)0x04
#define HALL_ERROR              (u32)0x08
#define OVER_VOLTAGE            (u32)0x10
#define MCU_ERROR               (u32)0x20
#define MOTOR_BLOCK             (u32)0x40
#define FOOTPLATE_ERR           (u32)0x80
#define SPEED_CONTROL           (u32)0x100
#define WRITING_EEPROM          (u32)0x200
#define START_UP_FAILURE        (u32)0x400
#define SPEED_FEEDBACK          (u32)0x800
#define OVERHEAT                (u32)0x1000
#define OVER_CURRENT1           (u32)0x2000
#define ACCELERATE_PADAL_ERR    (u32)0x4000
#define ICS1_ERR                (u32)0x8000
#define ICS2_ERR                (u32)0x10000
#define BREAK_ERR               (u32)0x20000
#define HALL_SEL_ERROR          (u32)0x40000
#define MOSFET_DRIVER_FAULT     (u32)0x80000
#define MOSFET_HIGH_SHORT       (u32)0x100000
#define PHASE_OPEN              (u32)0x200000
#define PHASE_SHORT             (u32)0x400000
#define MCU_CHIP_ERROR          (u32)0x800000
#define PRE_CHARGE_ERROR        (u32)0x1000000
#define OVERHEAT1               (u32)0x8000000
#define SOC_ZERO_ERROR          (u32)0x80000000
```

If the fault code is as above, please contact after-sales service for specific solutions

The controller is connected virtual, and the upper computer has no voltage display

Problem:The controller is connected to the host computer, but there is no voltage display on the Display interface



Reason: The old version of the host computer has a BUG, and there may be a virtual connection.

Solutions:

1.The host computer switches back to the PAGE1 interface, and click CONECT again

2.Restart the power and reconnect the host computer

3.Replace the host computer version EM_V3-EMsport-SIA-1

Can't reach maximum speed

Problem: Turn the handle to the maximum, the motor cannot reach the maximum speed

Reason:

1. If the handlebar has no maximum voltage output, there may be a hardware problem；There is a hardware problem, the customer cannot solve it by himself, and needs to be returned to the factory for repair；

2.Does not match the customer's motor before leaving the factory；
3.The customer changed the host computer settings；
4.The controller version, configuration, and program are incorrect；
Solutions:
1. Check whether the output voltage of the signal line of the switch can be above 3.5V；
2. Connect the host computer to check the controller,Set the enable value of downhill electric brake on PAGE2 interface，Is it the maximum speed the customer wants；
3. Check the motor setting interface of the PAGE3 interface,Is it correct, the parameters are in line with the current running motor；
4. Check the three-speed ratio setting of the PAGE2 interface；
5. View the current gear of the Display interface；

Abnormal communication and no response

Problem:

Solutions:
1. Check whether the controller is powered on；
2. Check whether the USB debugging cable is used correctly；（CAN is black, without CAN is blue）
3. Check the settings of the host computer: CAN needs to be checked, and no CAN needs to be checked；
4. Check whether the computer device manager port is consistent with the port displayed on the host computer；
5.If the operation is correct, please contact the after-sales department

Voltage and actual display are inaccurate

Problem:The host computer voltage and the actual battery voltage display are inaccurate
Reason:
1. Inaccurate calibration value before leaving the factory
2.The customer uses a host computer to connect two types of controllers at the same time, causing the configuration to be overwritten；
Solutions:

1. The controller voltage calibration value is wrong, you need to enter the advanced settings, click on the expansion interface to re-calibrate the voltage； After calibration, click on the calibration data to write in.；

Current and actual display are not accurate

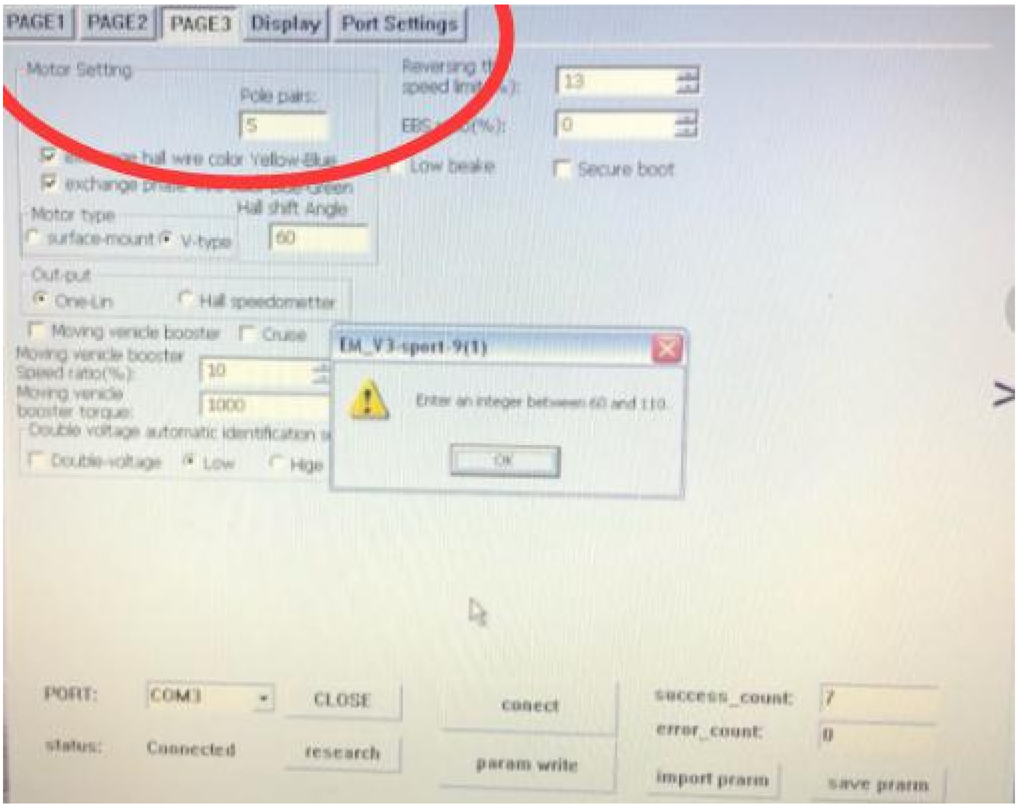
Problem: The display of the host computer current and the actual battery output current is not accurate
Reason:
1. Inaccurate calibration value before leaving the factory
2.The customer uses a host computer to connect two types of controllers at the same time, causing the configuration to be overwritten
Solutions:
1. Controller current calibration value is wrong, need to enter advanced settings， Click on the extended interface to re-calibrate the current; after calibration, click on the calibration data to write；

MOS tube breakdown

Problem: 1.The motor is connected to the controller, and the motor is pushed by the resistance by hand；
Reason:
1. The controller outputs a large current for a long time, causing the MOS tube to burst；
2. Controller positive and negative and connected reversely；
3. Controller overvoltage caused;（The controller exceeds its own voltage）
4.Battery BMS protection during riding, causing the controller to power off,The controller cannot protect itself. At this time, the faster the riding speed, the more likely the MOS tube will be damaged.（It is recommended that customers set the discharge current of the controller according to the peak discharge current of the battery BMS）
Solutions:
1. MOS breakdown customers cannot solve by themselves and need to return to the factory for repair；

Parameters cannot be modified

Problem: The upper computer parameters cannot be modified, and they will still be changed back to the original parameters after saving.； Prompt that the parameter value is wrong as shown in the figure below60-110



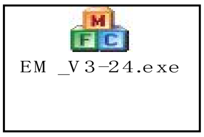
Reason:

1. Maybe the customer has changed some parameters, and the value does not meet the standard.;

2.The customer uses the wrong host computer version;

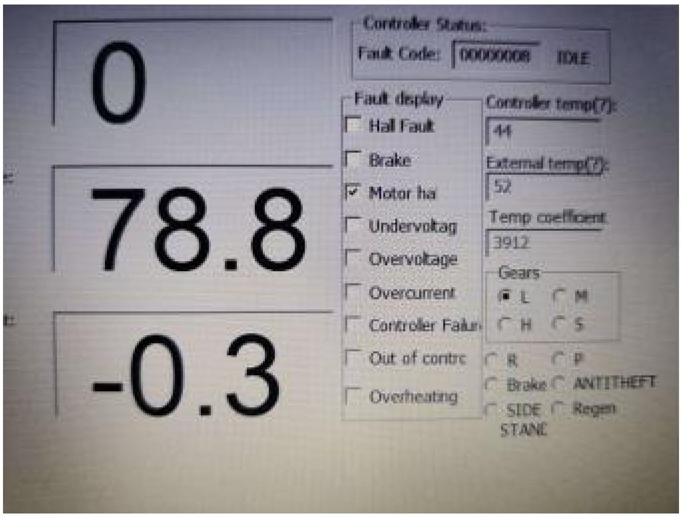
Solutions:

Updating the host computer version can solve this problem



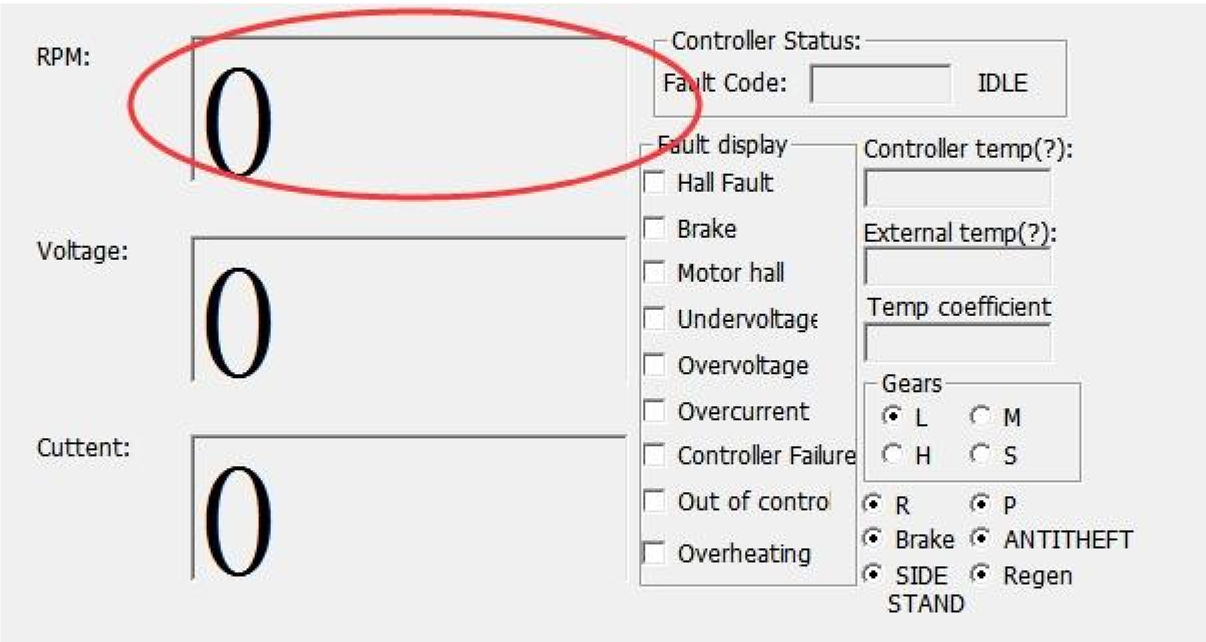
Controller Hall Error

Problem: Show Motor Hall error in display page



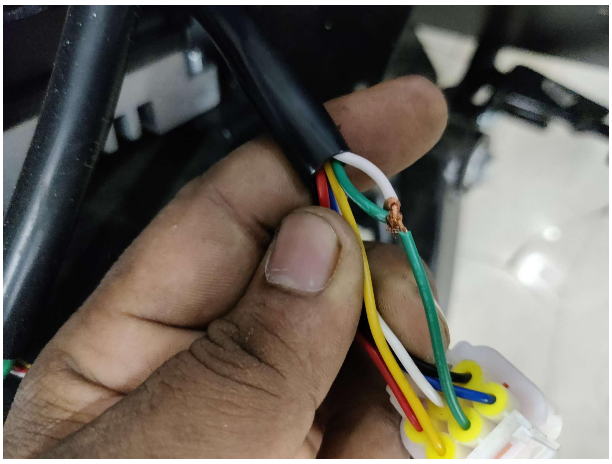
Analysis:

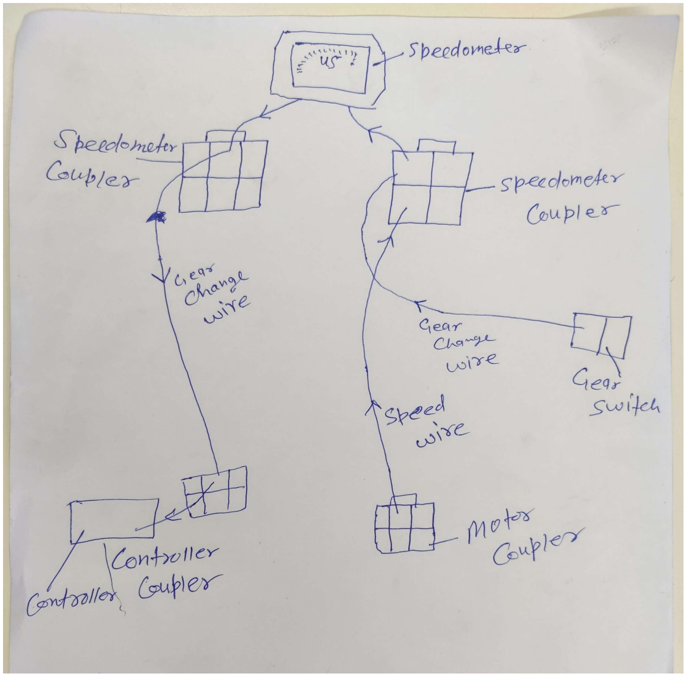
• When turn the motor by hand, and connect controller with PC.
will the RPM change?



• Make remote control and video call to figure out the problem
If problem not able to be solved, send back the controller for further analysis

Reason:
Controller MCU Hall failed, caused by connect green hall to speedometer from motor/controller hall set.





Solutions: Set controller one-line signal to hall signal, then connect with speedometer.